The coupled field equations of the 7+1 model and their solutions in the three gravity regimes: symbols and their definitions

A dictionary of every symbol used for the action term by term, the coupled field equations, the radion, and the accurate approximate solutions in the primordial, modern and far-future regimes — sections 20–23 of the notebook 7+1.nb

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
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Abstract

For each symbol this dictionary gives the mathematical notation, the name used in the Mathematica notebook 7+1.nb (where there is one), the defining equation and its meaning, grouped by topic. Longer defining equations follow each table as numbered displayed formulas. Every equation needed to define a symbol is written in this document.

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1. Conventions

- Units: $\hbar = c = 1$. Sections 21 and 22.1 of the notebook (regime I and the capture of the auxiliary space) use the Gauss–Bonnet length $\ell_2 = \sqrt{\alpha_2} = 1$ as unit of length and time; sections 22.2–23 use GeV, s, K, Gyr where stated.
- Coordinates $X = (x_0, x_1, \ldots, x_7)$: x_0 is time (the direction of the octonion unit $e_0 = 1$), x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ the four auxiliary space-like coordinates; on the world brane $x = (x_0, x_1, x_2, x_3)$, and $y = (x_4, \ldots, x_7)$.
- Indices: $\mu, \nu, \rho, \sigma, \lambda, \alpha, \beta, \gamma, \delta = 0, \ldots, 7$ coordinate indices (on the brane $0, \ldots, 3$); $A, B, C = 0, \ldots, 7$ frame indices; $k, l = 4, \ldots, 7$ auxiliary directions; $i, j = 1, 2, 3$ ordinary directions; repeated indices are summed.
- A prime is d/dx_0; $\tau = x_0 - x_s$ is the time since the singularity x_s; "ln" is the natural logarithm.
- Signature: the bulk has one time-like and seven space-like dimensions; in xAct the list $\{7, 1, 0\}$ declares exactly this (seven space-like, one time-like, no degenerate dimension). The frame metric $\eta = \mathrm{diag}(+1, -1, \ldots, -1)$ (`η17`) is simply the squares of the octonion basis, $\eta_{AB} = \mathrm{Re}(e_A e_B)$.

The signature: $\{7,1,0\}$ and $\mathrm{diag}(+1,-1,\ldots,-1)$. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7,1,0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta_{71} = \mathrm{diag}(-1,+1,\ldots,+1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like = $+1$, time-like = -1). (ii) The matrix $\eta = -\eta_{71} = \mathrm{diag}(+1,-1,\ldots,-1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1, \ldots, e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \quad (i = 1,\ldots,7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

η gives the signs of the Clifford relation $\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\,\eta_{AB}\,1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p,q)$, with p generators squaring to $+1$ and q to -1, the algebra $Cl(1,7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same in both sign conventions, since $\det\eta = \det\eta_{71} = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7,1,0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$ and $\mathrm{Re}(x\,x) = \eta_{AB}x^A x^B$.

2. Spacetime, curvature and the Lovelock tensors

symbol	Mathematica	definition	meaning
$g_{\mu\nu}$, $g^{\mu\nu}$, $\lvert g\rvert$	`curvature[g]["g"]`, `["ginv"]`; `gCos` (cosmological)	$ds^2 = g_{\mu\nu}dX^\mu dX^\nu$	metric, inverse, $\lvert\det g\rvert$
$\{^{\lambda}_{\mu\nu}\}$	`curvature[g]["chr"]`	(2.1)	Christoffel symbols
$R^\rho{}_{\sigma\mu\nu}$, $R_{\sigma\nu}$, R	`["riem"]`, `["ric"]`, `["R"]` of `curvature[g]`	(2.1)	Riemann tensor, Ricci tensor, scalar curvature
$G^\mu{}_\nu$	`curvature[g]["G"]`	$R^\mu{}_\nu - \frac{1}{2}\delta^\mu_\nu R$	Einstein tensor
$\delta^{\mu_1\cdots\mu_n}_{\nu_1\cdots\nu_n}$	`kδ` (xAct), pair sets	$\det[\delta^{\mu_i}_{\nu_j}]$	generalized Kronecker delta
L_k	`lovelockL[ra, k]`	(2.2)	Lovelock density of order k ($L_0 = 1$, $L_1 = R$)
L_2	—	$R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$	Gauss–Bonnet density
L_3	`feL3coef` (coefficients)	(2.3)	cubic Lovelock density
$E^{(k)\mu}{}_\nu$	`lovelockE`, `ECos`	(2.2)	Lovelock tensor of order k ($E^{(0)} = -\delta/2$, $E^{(1)} = G$, $E^{(4)} \equiv 0$)
α_k	`alf[k]`	$\alpha_0 = 2\Lambda$, $\alpha_1 = 1$	Lovelock couplings (α_2: length^2, α_3: length^4)
Λ	`lamb`		bulk cosmological constant
κ	`kapS` (sections 20–23), `k8` (section 18)	$8\pi G_8 = M_8^{-6}$	eight-dimensional gravitational coupling
G_8, M_8	`rIIM8` ($M_8 = 10^4$ GeV)	$\kappa = 8\pi G_8 = M_8^{-6}$	Newton constant and reduced Planck mass in 8 dimensions
ℓ_2, ξ	`rIxi`, `xiS`	$\ell_2 = \sqrt{\alpha_2}$, $\xi = \alpha_3/\alpha_2^2$	Gauss–Bonnet length; cubic-to-Gauss–Bonnet ratio (10^{-6})
R_{abcd} (test)	`feRandCurv[seed]`	$\sum_{n=1}^{3}(S^{(n)}_{ac}S^{(n)}_{bd} - S^{(n)}_{ad}S^{(n)}_{bc})$	random integer algebraic curvature tensor ($S^{(n)}$ symmetric)

symbol	Mathematica	definition	meaning
invariants	`feInvariants[RL]`	(2.3)	the eight cubic invariants of a curvature tensor

$$
\begin{aligned}
\{^{\lambda}_{\mu\nu}\} &= \tfrac{1}{2} g^{\lambda\rho}(\partial_\mu g_{\rho\nu} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}), && (2.1)\\
R^{\rho}{}_{\sigma\mu\nu} &= \partial_\mu \{^{\rho}_{\sigma\nu}\} - \partial_\nu \{^{\rho}_{\sigma\mu}\} + \{^{\rho}_{\mu\lambda}\}\{^{\lambda}_{\sigma\nu}\} - \{^{\rho}_{\nu\lambda}\}\{^{\lambda}_{\sigma\mu}\},\\
R_{\sigma\nu} &= R^{\rho}{}_{\sigma\rho\nu}, \qquad R = g^{\sigma\nu} R_{\sigma\nu},\\
L_k &= \frac{1}{2^k} \delta^{\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\gamma_1\delta_1\cdots\gamma_k\delta_k} R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1} \cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k}, && (2.2)\\
E^{(k)\mu}{}_{\nu} &= -\frac{1}{2^{k+1}} \delta^{\mu\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\nu\gamma_1\delta_1\cdots\gamma_k\delta_k} R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1} \cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k},\\
L_3 &= R^3 - 12 R R_{\mu\nu} R^{\mu\nu} + 16 R^{\mu}{}_{\nu} R^{\nu}{}_{\rho} R^{\rho}{}_{\mu} + 24 R^{\mu\rho} R^{\nu\sigma} R_{\mu\nu\rho\sigma} && (2.3)\\
&\quad + 3 R R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} - 24 R^{\mu\nu} R_{\mu}{}^{\rho\sigma\tau} R_{\nu\rho\sigma\tau}\\
&\quad + 4 R^{\mu\nu}{}_{\rho\sigma} R^{\rho\sigma}{}_{\lambda\tau} R^{\lambda\tau}{}_{\mu\nu} - 8 R_{\mu}{}^{\rho}{}_{\nu}{}^{\sigma} R_{\rho}{}^{\lambda}{}_{\sigma}{}^{\tau} R_{\lambda}{}^{\mu}{}_{\tau}{}^{\nu}.
\end{aligned}
$$

3. The fields and the terms of the action

symbol	Mathematica	definition	meaning
$S_0, \ldots, S_3$	—	(3.1)	the four Lovelock terms of gravity
ψ, $\bar\psi$	`dirac16complex`	$\bar\psi = \psi^\dagger \Gamma_0$	the 16-component complex field and its Dirac conjugate
Γ_A, Γ^A	`Γ17[A]`, `Γ17up[A]`	(3.2)	real 16×16 Dirac matrices; $\Gamma^A = \eta^{AA}\Gamma_A$
τ_i^O	`τO[i]`	$(\tau_i^O)^2 = -1_8$, mutually anticommuting	real antisymmetric 8×8 matrices of the octonions
$e^A{}_\mu$, $e_A{}^\mu$	`frameData[cv, e]` (section 13)	$g_{\mu\nu} = \eta_{AB} e^A{}_\mu e^B{}_\nu$	frame (vielbein) and inverse frame
$\Gamma^\mu(X)$	—	$e_A{}^\mu \Gamma^A$	frame matrices
$\omega_\mu{}^A{}_B$, Ω_μ	—	(3.3)	spin connection and its spinor form
D_μ	—	$\partial_\mu + \Omega_\mu$ (bulk; brane fields: (3.6))	covariant derivative
m	`mPs` (section 18)		mass of dirac16complex
g_r	`gg[r]` (section 15)	$r = 0, \ldots, 8$	the nine self-couplings (dimension length^6)
K, Θ^K, η_K	—	(3.4)	multi-index, Clifford product, its sign
Θ_K	—	$\eta_K \Theta^K$	lowered Clifford product
$U(b)$, U_0, K_U, b_0	`uPot`, `u0B`, `kU`	(3.5)	stabilizing potential of the auxiliary space, its minimum value, stiffness, minimum
W, $h_{\mu\nu}$	—	$W = \{y = 0\}$, $h_{\mu\nu} = g_{\mu\nu}(x, 0)$	world brane and its induced metric
T^4, L_c	`lcT`	$x_k \simeq x_k + L_c$	auxiliary torus and its coordinate period
f_W	`fWb`		brane tension (energy per brane volume)
$\mathcal{L}_{\rm SM}$	—	(3.6)	Standard Model Lagrangian on the brane
G^a_μ, W^I_μ, B_μ	—	(3.6)	gauge fields of $SU(3)$, $SU(2)$, $U(1)_Y$

symbol	Mathematica	definition	meaning
g_s, g, g'	—		gauge couplings
T^a, f^{abc}, τ^I, ϵ^{IJK}	—	$[T^a, T^b] = if^{abc}T^c$	generators and structure constants
Φ, $\tilde{\Phi}$	—	$\tilde{\Phi} = i\sigma_2\Phi^*$	Higgs doublet and its conjugate
μ^2, λ	—		Higgs potential parameters
Y_u, Y_d, Y_e, Y_ν	—		Yukawa matrices
Q_L, u_R, d_R, L_L, e_R, ν_R, f	—		quark and lepton fields; f any of them
y_{is}, G_P^f, λ_P	—	(3.7)	Yukawa, vector and Higgs portal couplings
ψ_s, χ_s, P_R	—	$P_R = \frac{1}{2}(1+\gamma_5)$	brane-right-handed parts of dirac16complex at the brane; portal spinors
ϕ	—	$\sqrt{12}\,M_{\rm Pl}\ln(b/b_0)$	canonical radion
$\mathcal{L}_{\phi W}$	—	(3.8)	radion–brane coupling
$b_{\rm QCD}$, n_f, α_s	—	$b_{\rm QCD} = 11 - \frac{2}{3}n_f$	first coefficient of the QCD beta function, active quark flavours, strong coupling

$$S_k = -\frac{\alpha_k}{2\kappa}\int d^8X\sqrt{|g|}\,L_k \quad (k=0,1,2,3), \tag{3.1}$$

$$S_\psi = \int d^8X\sqrt{|g|}\,\bar\psi\, i\Gamma^\mu(\partial_\mu+\Omega_\mu)\psi, \qquad S_m = -\int d^8X\sqrt{|g|}\,m\bar\psi\psi,$$

$$S_{\rm self} = -\frac{1}{2}\sum_{r=0}^{8} g_r\int d^8X\sqrt{|g|}\sum_{|K|=r}\eta_K(\bar\psi\Theta^K\psi)^2, \qquad S_U = -\int d^8X\sqrt{|g|}\,U(b),$$

$$S_f = -\int_W d^4x\sqrt{|h|}\,f_W, \qquad S_{\rm SM} = \int_W d^4x\sqrt{|h|}\,\mathcal{L}_{\rm SM},$$

$$\int_W d^4x\sqrt{|h|}(\dots) = \int d^8X\sqrt{|g|}\,\frac{\delta^4(y)}{b^4}(\dots),$$

$$\Gamma_0 = \begin{pmatrix}0 & 1_8\\ 1_8 & 0\end{pmatrix}, \quad \Gamma_i = \begin{pmatrix}\tau_i^O & 0\\ 0 & -\tau_i^O\end{pmatrix}, \quad \Gamma_A\Gamma_B+\Gamma_B\Gamma_A = 2\eta_{AB}1_{16}, \tag{3.2}$$

$$\omega_\mu{}^A{}_B = e^A{}_\nu\big(\partial_\mu e_B{}^\nu + \{{}^\nu_{\mu\lambda}\}e_B{}^\lambda\big), \qquad \Omega_\mu = \tfrac{1}{4}\eta_{AC}\,\omega_\mu{}^C{}_B\,\Gamma^A\Gamma^B, \tag{3.3}$$

$$\Theta^K = i^{\sigma_r}\Gamma^{A_1}\cdots\Gamma^{A_r}, \quad K=(A_1<\cdots<A_r), \quad \sigma_r = \tfrac{r(r-1)}{2} \bmod 2, \tag{3.4}$$

$$\eta_K = \eta_{A_1A_1}\cdots\eta_{A_rA_r}, \qquad \Theta^{()} = 1_{16},$$

$$U(b) = U_0 + \frac{K_U}{2}\Big(\ln\frac{b}{b_0}\Big)^2, \qquad b^8 = \det(-g_{kl}), \tag{3.5}$$

$$\begin{aligned}\mathcal{L}_{\rm SM} = {}& -\tfrac{1}{4}G^a_{\mu\nu}G^{a\mu\nu} - \tfrac{1}{4}W^I_{\mu\nu}W^{I\mu\nu} - \tfrac{1}{4}B_{\mu\nu}B^{\mu\nu} + (D_\mu\Phi)^\dagger D^\mu\Phi\\ & - \mu^2\Phi^\dagger\Phi - \lambda(\Phi^\dagger\Phi)^2 + \sum_f \bar f i\gamma^\mu D_\mu f\\ & - \big[\bar Q_L Y_u\tilde\Phi u_R + \bar Q_L Y_d\Phi d_R + \bar L_L Y_e\Phi e_R + \bar L_L Y_\nu\tilde\Phi\nu_R + \text{h.c.}\big],\end{aligned} \tag{3.6}$$

$$D_\mu = \nabla_\mu - ig_sG^a_\mu T^a - igW^I_\mu\tfrac{\tau^I}{2} - ig'YB_\mu,$$

$$G^a_{\mu\nu} = \partial_\mu G^a_\nu - \partial_\nu G^a_\mu + g_sf^{abc}G^b_\mu G^c_\nu, \qquad W^I_{\mu\nu} = \partial_\mu W^I_\nu - \partial_\nu W^I_\mu + g\epsilon^{IJK}W^J_\mu W^K_\nu,$$

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu,$$

$$\begin{aligned}S_P = \int_W d^4x\sqrt{|h|}\Big[& -\sum_{i,s} y_{is}\bar L_i\tilde\Phi P_R\psi_s + \text{h.c.}\\ & - G_P^f(\bar\psi\Gamma^\mu\psi)|_W(\bar f\gamma_\mu f) - \lambda_P\Phi^\dagger\Phi(\bar\psi\psi)|_W\Big],\end{aligned} \tag{3.7}$$

$$\mathcal{L}_{\phi W} = \frac{\phi}{\sqrt{3}M_{\rm Pl}} T^{\mu}{}_{\mu}\big|_{\rm brane}, \qquad T^{\mu}{}_{\mu} \supset m_f \bar{f} f, \qquad T^{\mu}{}_{\mu} \supset -\frac{b_{\rm QCD}\alpha_s}{8\pi} G^a_{\mu\nu} G^{a\mu\nu}. \tag{3.8}$$

4. Energy–momentum tensors and the homogeneous metric

symbol	Mathematica	definition	meaning
$T_{\mu\nu}$	—	$\frac{2}{\sqrt{\lvert g\rvert}}\frac{\delta S}{\delta g^{\mu\nu}}$	energy–momentum tensor of a sector ($T^0{}_0 = \varepsilon$)
T_ψ, T_U, T_f, $T_{\rm SM}$, T_P	—		energy–momentum tensors of the sectors
a, b, n	`aS`, `bS`; `a3`, `b4`	(4.1)	scale factors of ordinary and auxiliary space; lapse ($n = 1$)
H_a, H_b	`Ha`, `Hb`	a'/a, b'/b	Hubble rates
H_a', H_b'	`dHa`, `dHb`	dH_a/dx_0, dH_b/dx_0	their derivatives
ε, p_a, p_b	`epsM`, `paM`, `pbM`	$T^{\mu}{}_{\nu} = \mathrm{diag}(\varepsilon, -p_a \times 3, -p_b \times 4)$	energy density, pressures along ordinary and auxiliary space
$W(a,b)$	`wW[a, b]`	$a^3 b^4 \varepsilon$	comoving energy at fixed occupation numbers
k_a, k_b	`kaP`, `kbP`		comoving momentum of a particle along ordinary/auxiliary space
ω	`root` in `rIgas`	(4.3)	energy of a massless particle (times a^3b^4/V)
$V_\perp$	`vPerpS`, `rIIVp`	$b^4 L_c^4$	proper volume of the auxiliary torus
ρ_4, p_4, w	`rho4`, `wW0`	$p_4 = w\rho_4$	energy density and pressure of a brane fluid
θ, $\langle\cdot\rangle$	`rIth`, `rIwt`	(4.3)	momentum angle and the average over the momentum sphere
C_g	`rICg`	3×10^{-6} (units $\ell_2 = 1$)	normalization of the dirac16complex gas
$\mathrm{EL}_n, \mathrm{EL}_a, \mathrm{EL}_b$	`feELn`, `feELa`, `feELb`	Euler–Lagrange expressions	minisuperspace variations
e_{00}, e_{11}, e_{44}	`eTot[1]`, `eTot[2]`, `eTot[5]`; `feTot[c]` ($\alpha_1 = 1$)	$\sum_k \alpha_k E^{(k)c}{}_c - \Lambda$	left-hand sides of the 00, 11, 44 equations

$$ds^2 = n^2 dx_0^2 - a^2(dx_1^2 + dx_2^2 + dx_3^2) - b^2(dx_4^2 + \cdots + dx_7^2), \tag{4.1}$$

$$\varepsilon = \frac{W}{a^3 b^4}, \qquad p_a = -\frac{\partial_a W}{3a^2 b^4}, \qquad p_b = -\frac{\partial_b W}{4a^3 b^3}, \tag{4.2}$$

$$0 = \varepsilon' + 3H_a(\varepsilon + p_a) + 4H_b(\varepsilon + p_b),$$

$$\kappa\varepsilon = \frac{C_g}{a^3 b^4}\langle\omega\rangle, \qquad \kappa p_a = \frac{C_g}{a^3 b^4}\Big\langle \frac{\cos^2\theta}{3a^2\omega}\Big\rangle, \qquad \kappa p_b = \frac{C_g}{a^3 b^4}\Big\langle \frac{\sin^2\theta}{4b^2\omega}\Big\rangle, \tag{4.3}$$

$$\omega = \sqrt{\frac{\cos^2\theta}{a^2} + \frac{\sin^2\theta}{b^2}}, \qquad \langle f\rangle = \frac{\int_0^{\pi/2} f \cos^2\theta \sin^3\theta \, d\theta}{\int_0^{\pi/2} \cos^2\theta \sin^3\theta \, d\theta}.$$

The field equations of the homogeneous metric (00, 11, 44 components) and the 00 equation written out:

$$\sum_k \alpha_k E^{(k)0}{}_0 = \kappa\varepsilon, \qquad \sum_k \alpha_k E^{(k)1}{}_1 = -\kappa p_a, \qquad \sum_k \alpha_k E^{(k)4}{}_4 = -\kappa p_b, \tag{4.4}$$

$$\begin{aligned}\kappa\varepsilon = {} & 3(H_a^2 + 4H_a H_b + 2H_b^2) - 12\alpha_2 H_b(4H_a^3 + 18H_a^2 H_b + 12H_a H_b^2 + H_b^3) \\ & + 360\alpha_3 H_a^2 H_b^3(4H_a + 3H_b) - \Lambda.\end{aligned}$$

5. The radion and the four-dimensional constants

symbol	Mathematica	definition	meaning
β	`betR`; `lb - rIIlnb0`	$\ln(b/b_0)$	radion (logarithmic size of the auxiliary space), $H_b = \beta'$
m_r	`rIIm` ($10^{-3}/\ell_2$), `rIIImr`	$m_r^2 = \kappa K_U/12$	radion mass
$\beta_{\rm eq}$	`bqV`	(5.2)	equilibrium of the radion
G_4, Λ_4	—	(5.1)	four-dimensional Newton constant and cosmological constant
$M_{\rm Pl}$	`rIIMPl`	2.435×10^{18} GeV, $M_{\rm Pl}^2 = M_8^6 V_\perp$	reduced four-dimensional Planck mass
a_E, H_E	`rIIHE`	$a_E = ab^2$, $H_E = H_a + 2H_b$	Einstein-frame scale factor and expansion rate
g_E	—	$h = (b_0/b)^4 g_E$	Einstein-frame metric of the brane
$\ell_\perp$	`rIIVp^(1/4)`	$V_\perp^{1/4}$	size of the auxiliary torus (3.08×10^{-13} m)
m_f, N_c	—		fermion mass, number of colours
Γ, $T_{\rm RH}$	`rIIgam`, `rIITRH`	(5.3)	radion decay rate, reheating temperature
$N_{\rm KK}$	`rIINKK`	$V_\perp \frac{\pi^2}{2}\frac{(m_r/2)^4}{(2\pi)^4}$	Kaluza–Klein momenta below $m_r/2$
T_*	—	$(M_8^6/M_{\rm Pl})^{1/5}$	normalcy temperature (estimate)

$$3H_a^2 = 8\pi G_4\rho_4 + \Lambda_4, \qquad H_a' = -\frac{\kappa}{2}(\varepsilon + p_a), \tag{5.1}$$

$$8\pi G_4 = \frac{\kappa}{V_\perp}, \qquad \Lambda_4 = \Lambda + \kappa U_0 + \frac{\kappa f_W}{V_\perp},$$

$$\beta'' + 3H_a\beta' + 4\beta'^2 = \kappa\Big(\frac{\varepsilon}{6} - \frac{p_a}{2} + \frac{p_b}{3}\Big) + \frac{\Lambda}{3} \quad (\alpha_2 = \alpha_3 = 0), \qquad 3H_E^2 = 6\beta'^2 + \kappa\varepsilon,$$

$$\frac{\kappa}{4} bU'(b) = \Lambda + \kappa U + \frac{2\kappa f_W}{V_\perp} + \frac{\kappa}{2}(\varepsilon - 3p_a) + \kappa p_b - 12\alpha_2 H_a^2(H_a' + H_a^2) \quad \text{(static } b\text{)}, \tag{5.2}$$

$$\beta_{\rm eq} = \frac{[\dots]}{3m_r^2},$$

$$\Gamma(\phi \to f\bar f) = \frac{N_c m_f^2 m_r}{24\pi M_{\rm Pl}^2}\Big(1 - \frac{4m_f^2}{m_r^2}\Big)^{3/2}, \qquad \Gamma(\phi \to gg) = \frac{b_{\rm QCD}^2 \alpha_s^2 m_r^3}{96\pi^3 M_{\rm Pl}^2}, \tag{5.3}$$

$$T_{\rm RH} = \Big[\frac{\Gamma M_{\rm Pl}}{(\pi^2 g_*/90)^{1/2}}\Big]^{1/2}.$$

In (5.2) $[\dots]$ is the right-hand side of the equation for $\frac{\kappa}{4}bU'(b)$ with U replaced by U_0; with $U = U_0 + \frac{K_U}{2}\beta^2$ one has $bU' = K_U\beta$ and $\frac{\kappa}{4}K_U = 3m_r^2$.

6. Regime I: the primordial Lovelock regime

symbol	Mathematica	definition	meaning
τ, x_s	`tS`; `rIxs`	$\tau = x_0 - x_s$	time since the singularity; its position
u, X	`uS`, `xS`	$u = H_b/H_a$, $X = H_a^2$	variables of the vacuum curve
P_E, P_G, P_C	`rIpE`, `rIpG`, `rIpC`	(6.1)	Einstein, Gauss–Bonnet, cubic parts of the vacuum constraint
$X(u)$	`rIX[u, xi]`	(6.1)	the vacuum curve
F	`rIF[u, xi]`	$-H_a'$ on the curve	rate along the curve
$p_{\rm eff}$	`rIpeff[u, xi]`, `"peff"`	$-H_a^2/H_a'$	effective Kasner exponent ($= p$ for $a \propto \tau^p$)
u_1, p_1, q_1	`rIu1`, `rIp1`, `rIq1`	(6.2)	Einstein–Kasner constants
u_2, p_2, q_2	`rIu2`, `rIp2`, `rIq2`	(6.2)	Gauss–Bonnet–Kasner constants
h_0, h_1, h_2, A_0, A_1	`rIcub`; `ch1`, `ch2`, `cA0`, `cA1`	(6.3)	coefficients of the cubic-regime series
d_1, e_1	`rId1`, `rIe1`	(6.4)	Einstein-regime coefficients
f_1, f_2	`rIf1`, `rIf2`	$p_{\rm eff} = p_2 + f_1/H_a^2 + f_2\xi H_a^2$	Gauss–Bonnet-regime coefficients
s_1, s_2	`rIs1v`, `rIs2v`	(6.4), (6.5)	time shifts of the Einstein and Gauss–Bonnet formulas
$H_{a,\rm mid}$	`rIHmid`	$(f_1/(f_2\xi))^{1/4}$	middle of the Gauss–Bonnet regime
$\tau(u_2)$, $H_a(u_2)$, J	`rItauU2`, `rIHu2`, `rIJ`	(6.4)	quadrature data
τ_{32}, τ_{21}	—	$\lvert q_2\rvert\sqrt{30\xi}\,\ell_2$, ℓ_2	crossover times
h_*	`rIhStar`	$-\sqrt{\alpha_2/(54\alpha_3)}$	auxiliary contraction rate with matter near $\tau = 0$
w_a, w_b	`wA`	$p_a = w_a\varepsilon$, $p_b = w_b\varepsilon$	equations of state
P, Q, $\mathcal{C}$	`pP`, `qQ`, `cC`	$a \propto \tau^P$, $b \propto \tau^Q$, $\mathcal{C} = \kappa\varepsilon\tau^2$	power laws of Einstein gravity with a fluid
S_1, S_2, S_3	`"S"` in `rIstate`	(6.6)	sizes of the curvature terms of the 00 equation
$f_{\rm gas}$	`"frac"`	(6.6)	gas fraction
K, $F(K)$	`rIbog[k, p, q]`, `rIFk`	$\lvert\beta_K\rvert^2$	scaling variable of a mode; Bogoliubov coefficient (pair creation)
σ_1, σ_2	`rIsg1`, `rIsg2`	Pauli matrices	mode Hamiltonian $K\tau^{-p_1}\sigma_1 + K\tau^{-q_1}\sigma_2$
σ	`sV`	$-\ln H_a$	integration variable

$$P_E = 3(2u^2+4u+1), \quad P_G = -12u(u^3+12u^2+18u+4), \quad P_C = 360u^3(3u+4), \tag{6.1}$$
$$X(u) = \frac{P_G + \sqrt{P_G^2 - 4\xi P_C P_E}}{2\xi|P_C|},$$

$$u_1 = -1 + \tfrac{1}{\sqrt{2}}, \quad p_1 = \frac{1}{3+4u_1} = \frac{1+2\sqrt{2}}{7}, \quad q_1 = u_1 p_1, \tag{6.2}$$
$$0 = u_2^3 + 12u_2^2 + 18u_2 + 4, \quad u_2 \in (-\tfrac{1}{3}, 0), \quad p_2 = \frac{3}{3+4u_2}, \quad q_2 = u_2 p_2,$$

$$H_a = \frac{1}{\tau} + A_0 + A_1\tau, \quad H_b = h_0 + h_1\tau + h_2\tau^2, \quad h_0 = -\sqrt{\frac{\alpha_2}{30\alpha_3}}, \quad h_1 = \frac{\alpha_2}{16\alpha_3} - \frac{1}{32\alpha_2}, \tag{6.3}$$
$$A_0 = \frac{\sqrt{30}(58\alpha_2^2 - 45\alpha_3)}{1920\alpha_2^{3/2}\alpha_3^{1/2}}, \qquad A_1 = -\frac{3647\alpha_2}{92160\alpha_3} - \frac{223}{6144\alpha_2} + \frac{645\alpha_3}{8192\alpha_2^3},$$

$$h_2 = \frac{\sqrt{30}(-1212\alpha_2^4 + 1180\alpha_2^2\alpha_3 + 225\alpha_3^2)}{307200\alpha_2^{5/2}\alpha_3^{3/2}},$$

$$u = u_1 + d_1 H_a^2,\ d_1 = \frac{\alpha_2(16-11\sqrt{2})}{4}, \quad p_{\rm eff} = p_1 + e_1 H_a^2,\ e_1 = \frac{\alpha_2(1+9\sqrt{2})}{7}, \tag{6.4}$$

$$\tau = \frac{p_1}{H_a} - e_1 H_a + s_1, \qquad s_1 = \tau(u_2) - \frac{p_1}{H_a(u_2)} + J,$$

$$J = \int_0^{H_a(u_2)} \Big(\frac{1}{F} - \frac{p_1}{H^2}\Big) dH, \qquad \tau(H_a) = \int_{H_a}^{\infty} \frac{dH}{F},$$

$$\tau = \frac{p_2}{H_a} + \frac{f_1}{3H_a^3} - f_2\xi H_a + s_2 \quad \text{(Gauss–Bonnet regime)}, \tag{6.5}$$

$$S_1 = 3(H_a^2 + 4H_a|H_b| + 2H_b^2), \qquad S_2 = 12\alpha_2|H_b|(4H_a^3 + 18H_a^2|H_b| + 12H_aH_b^2 + |H_b|^3), \tag{6.6}$$

$$S_3 = 360\alpha_3 H_a^2|H_b|^3(4H_a + 3|H_b|), \qquad f_{\rm gas} = \frac{\kappa\varepsilon}{S_1 + S_2 + S_3 + \kappa\varepsilon}.$$

7. Regime II: the modern regime

symbol	Mathematica	definition	meaning
τ_E, A, ϑ	`rIItEF`, `rIIaF`, `rIIphF`	(7.1)	Einstein-frame time origin, amplitude and phase of the radion oscillation
$\ln b_0$	`rIIlnb0`		minimum of U in the capture run
T, T_1	—	$T_1 = 10$ GeV	photon temperature; start of the $t(T)$ integration
g_*, g_{*s}	`rIIgstars[T]`	(7.2)	relativistic degrees of freedom for energy and entropy
g_i, m_i, x	`rIIsp`	$x = m_i/T$	degeneracy and mass of species i
K_1, K_2	`BesselK[1, …]`, `BesselK[2, …]`		modified Bessel functions of the second kind
s	—	$(2\pi^2/45)g_{*s}T^3$	entropy density
$S(T)$	`rIIS[T]`	$\frac{1}{2}[1 + \tanh((T - 0.160)/0.030)]$ (GeV)	QCD switch (quarks and gluons $\leftrightarrow$ hadrons)
T_ν, $N_{\rm eff}$, $g_{\gamma e}$	`rIIgge`	$T_\nu = T\min(1, (g_{\gamma e}/\frac{11}{2})^{1/3})$, $N_{\rm eff} = 3.044$	neutrino temperature, effective number, g_{*s} of γ and $e^\pm$
$t(T)$	`rIItg`	(7.3)	time–temperature relation of the radiation era
H_0, Ω_m, Ω_r, Ω_Λ, Ω_γ	`rIIH0s`, `rIIOm`, `rIIOr`, `rIIOL`, `rIIOg`	(7.4)	Planck 2018 parameters and the radiation densities
$T_{\rm CMB}$	`rIITcmb`	2.7255 K	temperature of the cosmic microwave background
t_{rm}, $t_{m\Lambda}$, t_m	`rIItrm`, `rIItmL`, `rIItm`	(7.4)	exact two-component solutions and the matter solution
$z_{\rm eq}$, t_0	`rIIage`	$\Omega_m/\Omega_r - 1$; $\int_0^1 \frac{da}{aH_a}$	redshift of matter–radiation equality, age

$$\beta \approx \frac{A\sin(m_r\tau + \vartheta)}{m_r(\tau - \tau_E)}, \qquad H_E \approx \frac{2}{3(\tau - \tau_E)}, \tag{7.1}$$

$$\rho_i = \frac{g_i T^4}{2\pi^2}\sum_{n\geq 1}(\pm 1)^{n+1}\frac{3(nx)^2K_2(nx) + (nx)^3K_1(nx)}{n^4}, \tag{7.2}$$

$$p_i = \frac{g_i T^4}{2\pi^2}\sum_{n\ge 1}(\pm 1)^{n+1}\frac{(nx)^2 K_2(nx)}{n^4}, \qquad g_* = \frac{30\rho}{\pi^2 T^4}, \qquad g_{*s} = \frac{45 s}{2\pi^2 T^3},$$

$$H_a = \Big(\frac{\pi^2 g_*}{90}\Big)^{1/2}\frac{T^2}{M_{\rm Pl}}, \qquad \ln a = -\ln T - \tfrac{1}{3}\ln g_{*s} + \text{const}, \tag{7.3}$$

$$t(T) = t(T_1) + \int \frac{d\ln a}{H_a}, \qquad t(T_1) = \frac{1}{2H_a(T_1)},$$

$$H_a = H_0\sqrt{\Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_\Lambda}, \qquad \Omega_\gamma = \frac{(\pi^2/15)T_{\rm CMB}^4}{3H_0^2 M_{\rm Pl}^2}, \tag{7.4}$$

$$\Omega_r = \Omega_\gamma\Big[1 + \frac{7}{8}\Big(\frac{4}{11}\Big)^{4/3} N_{\rm eff}\Big], \qquad t_m = \frac{2a^{3/2}}{3H_0\Omega_m^{1/2}},$$

$$t_{rm} = \frac{2}{3H_0\Omega_m^2}\Big[(\Omega_m a - 2\Omega_r)\sqrt{\Omega_r + \Omega_m a} + 2\Omega_r^{3/2}\Big], \qquad t_{m\Lambda} = \frac{2}{3H_0\Omega_\Lambda^{1/2}}\operatorname{arcsinh}\Big(\sqrt{\frac{\Omega_\Lambda}{\Omega_m}}a^{3/2}\Big).$$

In (7.2) the upper sign is for bosons, the lower (alternating) for fermions; $s_i = (\rho_i + p_i)/T$.

8. Regime III: the far future

symbol	Mathematica	definition	meaning
H_∞	`rIIIH`	$H_0\Omega_\Lambda^{1/2}$	final de Sitter expansion rate
$T_{\rm GH}$	—	$H_\infty/(2\pi)$	Gibbons–Hawking temperature
M	—		mass of a created particle
γ	`rIIIgam`	(8.1)	friction coefficient of the radion in de Sitter
$m_{\rm eff}$	`rIIImeff`	(8.1)	effective radion mass in de Sitter
H	`hS`, `h8`		de Sitter rate of the linearization; of eight-dimensional de Sitter
Λ, m_r (test)	`rIIIlam`, `rIIImr`	3×10^{-4}, 0.1 (units $\ell_2 = 1$)	parameters of the numerical de Sitter test

$$a(t) = \Big(\frac{\Omega_m}{\Omega_\Lambda}\Big)^{1/3}\sinh^{2/3}\Big(\frac{3}{2}H_\infty t\Big), \qquad H_a = H_\infty\coth\Big(\frac{3}{2}H_\infty t\Big), \qquad |\beta_M|^2 = \frac{1}{e^{2\pi M/H_\infty}+1},$$

$$0 = \beta'' + \gamma\beta' + m_{\rm eff}^2(\beta - \beta_{\rm eq}), \tag{8.1}$$

$$\frac{\gamma}{3H} = \frac{1 - 16\alpha_2 H^2 + 48\alpha_2^2 H^4 - 48\alpha_3 H^4 + 512\alpha_2^3 H^6 + 192\alpha_2\alpha_3 H^6 - 1024\alpha_2^4 H^8}{(1 - 4\alpha_2 H^2 + 32\alpha_2^2 H^4)^2},$$

$$\frac{m_{\rm eff}^2}{m_r^2} = \frac{1}{1 - 4\alpha_2 H^2 + 32\alpha_2^2 H^4}, \qquad 3m_r^2\beta_{\rm eq} = 3H^2 - 12\alpha_2 H^4,$$

$$\Lambda = 21H^2 - 420\alpha_2 H^4 + 2520\alpha_3 H^6 \quad (a = b = e^{Hx_0}).$$

9. Symbols that look alike

symbol	**not to be confused with**
$\beta = \ln(b/b_0)$ (radion)	$\lvert\beta_K\rvert^2 = F(K)$, $\lvert\beta_M\rvert^2$ (Bogoliubov coefficients); α_k, β_k (dummy indices of the generalized delta)
ϕ (canonical radion)	ϑ (phase of the radion oscillation); Φ (Higgs doublet); θ (momentum angle)
b (auxiliary scale factor), b_0 (minimum of U)	b_{QCD} (QCD beta-function coefficient); the b quark
Γ_A (16×16 Dirac matrices), $\Gamma^\mu(X)$ (frame matrices)	Γ (decay rate); $\{^{\lambda}_{\mu\nu}\}$ (Christoffel symbols)
γ (radion friction)	γ^μ, γ_5 (4×4 Dirac matrices of the brane)
$S_0, \ldots, S_3, S_\psi, \ldots$ (terms of the action)	S_1, S_2, S_3 (sizes of the curvature terms, (6.6)); $S(T)$ (QCD switch); s (entropy density); s_1, s_2 (time shifts)
W (world brane)	$W(a, b)$ (comoving energy); W^I_μ (electroweak gauge fields)
C_g (gas normalization)	$\mathcal{C} = \kappa\varepsilon\tau^2$ (power-law constant); N_c (colours)
F (rate $-H'_a$ along the vacuum curve)	$F(K)$ (Bogoliubov coefficient)
X (coordinates $x_0, \ldots, x_7$)	$X = H_a^2$ (variable of the vacuum curve)
η (frame metric: the squares of the octonion basis)	η_K (sign of a multi-index); $\{7, 1, 0\}$ (xAct signature list: seven space-like, one time-like dimension)
κ ($8\pi G_8$)	K (scaling variable of a mode, or a multi-index); K_U (stiffness of U); K_1, K_2 (Bessel functions)
p_1, p_2, q_1, q_2 (Kasner exponents)	p_a, p_b (pressures); p_{eff}; P, Q (fluid power laws); P_E, P_G, P_C (polynomials)
h_0 (vacuum cubic-regime rate)	h_* (with matter); $h_{\mu\nu}$ (induced brane metric)
Λ (bulk)	Λ_4 (four-dimensional); Ω_Λ
τ (time since the singularity)	τ_i^O (octonion matrices), τ^I (Pauli matrices of $SU(2)$), τ lepton; τ_E, τ_{32}, τ_{21}